Assignment 2 Explanation

Introduction to Database Systems

DataLab

CS, NTHU

Outline

- Client-side classes
  - *As2BenchmarkRte*
  - *As2UpdateItemPriceParamGen*
  - *UpdateItemPriceTxnJdbcJob*
- Server-side classes
  - *UpdateItemPriceProcParamHelper*
  - *UpdateItemPriceTxnProc*
- StatisticMgr

Outline

- Client-side classes
  - *As2BenchmarkRte*
    1. *Executor*
    2. *READ_WRITE_TX_RATE*
  - *As2UpdateItemPriceParamGen*
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Executor

Q: Which place is better to new an executor?

Solution :

```
protected As2BenchmarkTxExecutor getTxExeutor(As2BenchTransactionType type) {
    TxParamGenerator<As2BenchTransactionType> paraGen;
    switch (type) {
    case READ_ITEM:
        paraGen = new As2ReadItemParamGen();
        break;

    case UPDATE_ITEM_PRICE:
        paraGen = new As2UpdateItemPriceTxnParamGen();
        break;

    default:
        paraGen = new As2ReadItemParamGen();
        break;
    }
    executor = new As2BenchmarkTxExecutor(paraGen);
    return executor;
}
```

READ_WRITE_TX_RATE

Q: Use READ_WRITE_TX_RATE to control the UPDATE rate

Solution:

```
protected As2BenchTransactionType getNextTxType() {
    RandomValueGenerator rvg = new RandomValueGenerator();

    // flag would be 100 if READ_WRITE_TX_RATE is 1.0
    int flag = (int) (As2BenchConstants.READ_WRITE_TX_RATE * precision);

    if (rvg.number(min:0, precision - 1) < flag) {
        return As2BenchTransactionType.UPDATE_ITEM_PRICE;
    } else {
        return As2BenchTransactionType.READ_ITEM;
    }
}
```

FAQ:

Question 1: `READ_WRITE_TX_RATE` 的定義

> L 君: 請問調 ReadItemTxn 跟 UpdateItemPriceTxn 的 Ratio 是指說 1) 多個 RTE 跑不同 Txn , 2) 同一個 RTE 跑多個 Txn?

這個要求的意思是說，在一個 RTE 選擇要跑哪一種 txn (transaction 的縮寫) 時，有多少機率會選到

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    2. *Update return*
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Update SQL

Q: `sql = "UPDATE item SET i_price = i_price` ~~`+ " + pru + "WHERE i_id = " + iid;`~~

Parser:

```
private ModifyData modify() {
    lex.eatKeyword("update");
    String tblname = lex.eatId();
    lex.eatKeyword("set");
    Map<String, Expression> map = new HashMap<String, Expression>();
    while (lex.matchId()) {
        String fldname = id();
        lex.eatDelim('=');
        Expression newval = modifyExpression();
        map.put(fldname, newval);
        if (lex.matchDelim(','))
            lex.eatDelim(',');
    }
    Predicate pred = new Predicate();
    if (lex.matchKeyword("where")) {
        lex.eatKeyword("where");
        pred = predicate();
    }
    return new ModifyData(tblname, map, pred);
}
```

```
private Expression modifyExpression() {
    if (lex.matchKeyword("add")) {
        lex.eatKeyword("add");
        lex.eatDelim('(');
        Expression lhs = queryExpression();
        lex.eatDelim(',');
        Expression rhs = queryExpression();
        lex.eatDelim(')');
        return new BinaryArithmeticExpression(lhs, OP_ADD, rhs);
    } else if (lex.matchKeyword("sub")) {
        lex.eatKeyword("sub");
        lex.eatDelim('(');
        Expression lhs = queryExpression();
        lex.eatDelim(',');
        Expression rhs = queryExpression();
        lex.eatDelim(')');
        return new BinaryArithmeticExpression(lhs, OP_SUB, rhs);
    } else if (lex.matchKeyword("mul")) {
        lex.eatKeyword("mul");
        lex.eatDelim('(');
        Expression lhs = queryExpression();
        lex.eatDelim(',');
        Expression rhs = queryExpression();
        lex.eatDelim(')');
        return new BinaryArithmeticExpression(lhs, OP_MUL, rhs);
    } else if (lex.matchKeyword("div")) {
        lex.eatKeyword("div");
        lex.eatDelim('(');
        Expression lhs = queryExpression();
        lex.eatDelim(',');
        Expression rhs = queryExpression();
        lex.eatDelim(')');
        return new BinaryArithmeticExpression(lhs, OP_DIV, rhs);
    } else if (lex.matchId())
        return new FieldNameExpression(id());
    else
        return new ConstantExpression(constant());
}
```

Update return

Q: Update return should handle return that larger than 1.

Solution:

```
sql = "UPDATE item SET i_price = " + updatedPrice + " WHERE i_id = " + itemIds[i];

int result = statement.executeUpdate(sql);
if (result == 0) {
    throw new RuntimeException("cannot update the record with i_id = " + itemIds[i]);
}
```

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    1. *Reuse Helper*
  - *UpdateItemPriceTxnProc*
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Reuse Helper

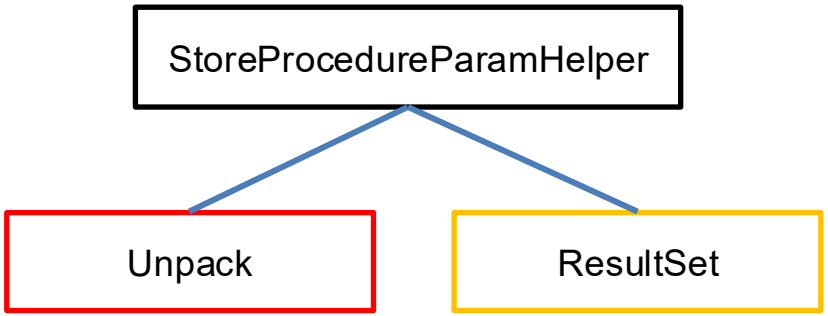
Q: Helper should be called in both JDBC and SP

1. ParamHelper is a server-side class

- org.vanilladb.bench.server.param.as2
  - ReadItemProcParamHelper.java
  - TestbedLoaderParamHelper.java
  - UpdateItemPriceProcParamHelper.java

2. ParamHelper also prepare resultset.

```
org.vanilladb.core.sql.storedprocedure
StoredProcedureParamHelper
  newDefaultParamHelper() : StoredProcedureParamHelper
  isReadOnly : boolean
  prepareParameters(Object...) : void
  getResultSetSchema() : Schema
  newResultSetRecord() : SpResultRecord
  setReadOnly(boolean) : void
  isReadOnly() : boolean
```



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    3. *VanillaDb.newPlanner()*
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VanillaDb.newPlanner()

Q: Instead of using StoreProcedureHelper.execute, use the title one

Solution:

```
Plan p = VanillaDb.newPlanner().createQueryPlan("SELECT i_name, i_price FROM item WHERE i_id = " + iid, tx);
Scan s = p.open();
```

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